Phase 1 Report

ProctorAI AI Based Online Proctoring System

This report defines the problem domain, reviews related work, checks project feasibility, and finalizes the datasets planned for the ProctorAI college project.

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1 Problem Domain and Problem Analysis

Even in colleges, similar training grounds, and recruitment the exam is conducted online. The primary challenge is that the candidate is not in a test centre/stage. An invigilator is able to see and hear the candidate in the physical examination, impose a search of the area, detect suspicious activity and act on it without delay. A candidate could switch test devices/allow another person to help during an exam, walk away, copy or view material on other tabs or deny the examiner permission to monitor at the right time, during an online test.

This project has an AI-based online examination proctoring problem domain. ProctorAI's solution is to track the candidate while the candidate takes a digital testing.ProctorAI's target is to watch a candidate during a test in the online application, and to uncover evidence of behavior that are deemed suspicious as integrity occasions. The system is not intended to fully replace a human proctor. Instead, it acts as a support to the proctor and capture the signals provided by the camera, microphone, screen recording, browser activity and device checks.

The project is important because, in my opinion, there are no ways to ensure exam integrity using only an online test page. A practical system requires set up before a test and subsequent monitoring during a test, and it requires no monitoring after a test.

Risk Area	Example Problem	ProctorAI Response
Candidate identity and presence	Candidate leaves the camera frame or another person appears.	Detect face absent and multiple person events.
Unauthorized devices	A mobile phone or second screen is used during the test.	Detect phone objects and extra display conditions.
Browser misuse	Candidate copies, pastes, exits fullscreen, or switches away from the test.	Track browser integrity events with evidence.
Environment audio	Background voice or suspicious noise is present.	Monitor microphone signals and report audio events.
Evidence review	Automatic detection may be wrong in real conditions.	Send evidence to the proctor for human review.

2 Literature Survey and Existing Systems

The current online proctoring services all take one of three forms, again: live proctoring, recorded proctoring, and automated proctoring using AI technology. Live proctoring requires that all candidates be directly watched by a proctor for their examinations. Video and activity logs stored by recorded proctoring for later review. Automated proctoring helps identify suspicious activity during an examination with the help of the artificial intelligence and browser signals.

There are some common AI-based proctoring techniques that have been studied such as log tracking, face detection, object detection, and behavior analysis. Cheating in an on-line test can happen in various ways, so these techniques can come in handy. However, camera monitoring can only detect the absence of a face, an absence of a mobile phone, and cannot do this by itself for copy paste activity. Another person will not be detected when using browser tracking metrics, but will be seen by the person using the tracking technologies. The multi signal approach is what makes ProctorAI possible.

In the studies conducted online, face detection and analysis of these faces are common topics. It is also crucial to detect objects, such as mobile phones or to see that there are two or more people there. This project uses YOLO as it is a real time object detection system, needed when the candidate is actively taking the exam. Furthermore, the COCO dataset is important since it is generally adopted in object detection and includes common categories of objects that are frequently utilized in comprehending the scene

Area Studied	What Existing Systems Use	Use in ProctorAI
AI proctoring	Face detection, object detection, log tracking, behavior analysis.	Used as the overall system direction.
Face monitoring	Face detection and recognition for identity and presence checks.	Used for face absent and multiple face events.
Object detection	Real time detection models such as YOLO.	Used for phone and person detection.
Browser monitoring	Activity logs and event tracking.	Used for copy paste, focus lost, fullscreen exit, and shortcuts.
Human review	Proctor reviews suspicious events and evidence.	Used to reduce unfair automatic decisions.

3 Proposed System Overview

ProctorAI is suggested as a web-based AI-based on-line proctoring system for college examinations. It includes two primary sections, one for candidates, and one for the examiner/ proctor. To select an assigned assessment, to complete setup check and take the test, the candidate portal is used. Assessments are created and candidates sessions monitored, integrity events received and stored in real-time on the proctor portal, and evidence is reviewed.

The system will deliver integrity events rather than outright make a candidate punishable. This is crucial since a light scene, camera positioning or background movement or normal user activity can trigger a false positive in any AI model. It ought to never be necessary for a human proctor to review evidence before making a decision.

Candidate portal: Assignment selection, Setup checks, Screen recording setup, Liveness verification, and Test attempt.

Proctor portal: Assessment creation, session monitoring, real time integrity event feed, evidence playback and event review.

AI service: face presence analysis, multiple person detection, phone detection and other camera based signals.

Monitoring the browser: copy paste, app focus lost, short cut attempts, screen recording status and exit full-screen mode

4 Feasibility Analysis

Technical feasibility: The project is technically feasible as the cameras, microphones and screen recording permissions are supported in most modern browsers. A back end API facilitates the management of Assessment, Session, Event, and Evidence. Real time communication can be used to send integrity events to the proctor portal. Real time object detection can be achieved using AI model like YOLO.

Environmental security: The system is elegant and feasible for college examinations, and allows candidates to check their system before taking a test, and proctors to monitor and analyze possible suspicious activities in the process of taking a test. This minimises the need for continual manual supervision, but does still allow human judgment to be included.

Economic viability: There is no cost of implementation as the project can be developed utilizing open source tools, browser APIs and publicly accessible datasets. The key expense for a college project is the time spent developing that project. If used for the public, the primary expenses would be storage for evidence clips and hosting.

Limitations: It is not possible to record the entire screen silently without the user's consent using Browser based screen sending recording. The false positive and false negative issue may occur with AI detection as well. For this reason, ProctorAI should enable, rather than enforce, human teachers to approve or punish, rather than do so automatically

5 Dataset Finalization

The following datasets and data sources are planned for ProctorAI. The goal is to use standard datasets for model support and a small custom validation set for realistic exam scenarios.

Dataset or Source	Purpose	Why It Is Suitable	Final Decision
COCO dataset	Object detection base data.	Contains labeled common objects and is widely used for object detection.	Use as base/reference dataset.
YOLO pretrained model data	Real time object detection.	Useful for detecting objects such as persons and phones during a live exam.	Use pretrained model and tune if needed.
Custom validation dataset	Project-specific testing.	Covers exam-like cases such as face absent, phone visible, multiple persons, poor lighting, and screen recording interruption.	Prepare a small validation set.
Browser event logs	Integrity event validation.	Captures copy paste, focus lost, fullscreen exit, and shortcut activity.	Generate during system testing.

6 Conclusion

The Phase 1 study results show that the project domain is relevant and feasible. An online examination environment can't be limited to a simple interface as candidates have the potential to use an unapproved device, deactivate the camera, take readymade help, or perhaps exploit browser features. Other studies reveal that face detection, object detection, log tracking and behavioral analysis are widely employed in online proctors.

ProctorAI will employ a multiple signal strategy and incorporate camera monitoring, microphone monitoring, screen monitoring, browser activity monitoring, connected display verification, liveness verification, and real-time event reporting. The system will report integrity events and give the proctor an opportunity to check the events report before deciding the final integrity of the system.

The finalized data sets and data sources also include the COCO data, data from the YOLO pretrained models, a custom validation model, and browser event logs obtained during the testing. Once problem analysis, literature survey and feasibility study is conducted, the project moves on to design and implementation phase.

References

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